

Solution. $P(Y=2) = P\{(1,2) \cup (2,2) \cup (3,2) \cup (4,2)\} = \frac{4}{10} = \frac{2}{5}$.

$$P(X=4) = P\{(4,1) \cup (4,2) \cup (4,3)\} = \frac{1}{10} + \frac{1}{10} + \frac{1}{20} = \frac{1}{4}$$

$$P(X=4, Y=2) = P\{(4,2)\} = \frac{1}{10}$$

$$P(Y=2 | X=4) = \frac{P(X=4 \cap Y=2)}{P(X=4)} = \frac{1/10}{1/4} = \frac{2}{5}$$

Now $P(X=4) \cdot P(Y=2) = \frac{1}{4} \cdot \frac{2}{5} = \frac{1}{10} = P(X=4 \cap Y=2)$

Hence the events $X=4$ and $Y=2$ are independent.

Problem 5.32. The joint probability distribution of two random variables X and Y given by: $P(X=0, Y=1) = \frac{1}{3}$, $P(X=1, Y=-1) = \frac{1}{3}$, and $P(X=1, Y=1) = \frac{1}{3}$.

Find (i) Marginal distributions of X and Y , and (ii) the conditional probability distribution of X given $Y=1$.

Solution. $P(X=-1)$

$$= \sum_y P(X=-1, Y=y)$$

$$= P(X=-1, Y=-1)$$

$$+ P(X=-1, Y=0)$$

$$+ P(X=-1, Y=1) = 0$$

$$\text{Similarly } P(X=0) = \frac{1}{3}$$

and $P(X=1) = \frac{2}{3}$

Thus

Marginal distribution of X is:

Values of X, x :	-1	0	1
$P(X=x)$:	0	$\frac{1}{3}$	$\frac{2}{3}$

Marginal distribution of Y is:

Values of Y, y :	-1	0	1
$P(Y=y)$:	$\frac{1}{3}$	0	$\frac{2}{3}$

(ii) The conditional probability distribution of X given Y is:

$$P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$P(X=-1 | Y=1) = \frac{P(X=-1, Y=1)}{P(Y=1)} = 0, P(X=0 | Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{1/3}{2/3} = \frac{1}{2}$$

$$P(X=1 | Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{1/3}{2/3} = \frac{1}{2}$$

Thus the conditional distribution of X given $Y=1$ is:

Values of $X=x$	-1	0	1
$P(X=x Y=1)$	0	$\frac{1}{2}$	$\frac{1}{2}$

Example 5.33. For the adjoining bivariate probability distribution of X and Y , find:

(i) $P(X \leq 1, Y=2)$,

(ii) $P(X \leq 1)$,

(iii) $P(Y \leq 3)$, and

(iv) $P(X < 3, Y \leq 4)$.

Y	1	2	3	4	5	6
X						
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Solution. The marginal distributions are given below:

Y	1	2	3	4	5	6	$p_X(x)$
X							
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
$p_Y(y)$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	$\sum p(x) = 1$ $\sum p(y) = 1$

$$(i) P(X \leq 1, Y=2) = P(X=0, Y=2) + P(X=1, Y=2) = 0 + \frac{1}{16} = \frac{1}{16}$$

$$(ii) P(X \leq 1) = P(X=0) + P(X=1) = \frac{8}{32} + \frac{10}{16} = \frac{7}{8}$$

$$(iii) P(Y \leq 3) = P(Y=1) + P(Y=2) + P(Y=3) = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

$$(iv) P(X < 3, Y \leq 4) = P(X=0, Y \leq 4) + P(X=1, Y \leq 4) + P(X=2, Y \leq 4)$$

$$= \left(\frac{1}{32} + \frac{2}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64}\right) = \frac{9}{16}$$

Example 5.34. For the joint probability distribution of two random variables X and Y given below:

Y	1	2	3	4	Total
X					
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
Total	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Find (i) the marginal distributions of X and Y , and
(ii) conditional distribution of X given the value of $Y=1$ and that of Y given the value of $X=2$.

Solution. The marginal distribution of X is defined as:

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$$\begin{aligned}\therefore P(X=1) &= \sum_y P(X=1, Y=y) \\ &= P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) + P(X=1, Y=4) \\ &= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}.\end{aligned}$$

$$\text{Similarly } P(X=2) = \sum_y P(X=2, Y=y) = \frac{9}{36}; P(X=3) = \sum_y P(X=3, Y=y) = \frac{7}{36}$$

$$\text{and } P(X=4) = \sum_y P(X=4, Y=y) = \frac{9}{36}.$$

Similarly, we can obtain the marginal distribution of Y.

MARGINAL DISTRIBUTION OF X

Values of X, x	1	2	3	4
P(X=x)	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

MARGINAL DISTRIBUTION OF Y

Values of Y, y	1	2	3	4
P(Y=y)	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

(ii) The conditional probability function of X given Y is defined as follows:

$$P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)} \text{ . Therefore}$$

$$\therefore P(X=1 | Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{4/36}{11/36} = \frac{4}{11}$$

$$P(X=2 | Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{1/36}{11/36} = \frac{1}{11}$$

$$P(X=3 | Y=1) = \frac{P(X=3, Y=1)}{P(Y=1)} = \frac{5/36}{11/36} = \frac{5}{11}$$

$$P(X=4 | Y=1) = \frac{P(X=4, Y=1)}{P(Y=1)} = \frac{1/36}{11/36} = \frac{1}{11}$$

Hence the conditional distribution of X given Y=1 is:

$$\begin{array}{c|cccc} x & 1 & 2 & 3 & 4 \\ \hline P(X=x | Y=1) & \frac{4}{11} & \frac{1}{11} & \frac{5}{11} & \frac{1}{11} \end{array}$$

Similarly, we can obtain the conditional distribution of Y for X=2 as given below:

$$\begin{array}{c|cccc} y & 1 & 2 & 3 & 4 \\ \hline P(Y=y | X=2) & \frac{1}{9} & \frac{1}{3} & \frac{1}{3} & \frac{2}{9} \end{array}$$

Example 5.35. A two-dimensional r.v. (X, Y) have a bivariate distribution given by:

$$P(X=x, Y=y) = \frac{x^2+y}{32}, \text{ for } x=0, 1, 2, 3 \text{ and } y=0, 1.$$

Find the marginal distributions of X and Y.

(b) A two-dimensional r.v. (X, Y) have a joint probability mass function:

$$p(x, y) = \frac{1}{27}(2x+y), \text{ where } x \text{ and } y \text{ can assume only the integer values } 0, 1 \text{ and } 2.$$

Find the conditional distribution of Y for X=x.

Solution. (a) We have

X	0	1	2	3	Marginal distribution of Y, P(Y=y)
Y					
0	0	$\frac{1}{32}$	$\frac{4}{32}$	$\frac{9}{32}$	$\frac{14}{32}$
1	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{18}{32}$
Marginal distribution of X, P(X=x)	$\frac{1}{32}$	$\frac{3}{32}$	$\frac{9}{32}$	$\frac{19}{32}$	1

The marginal probability distribution of X is given by:

$P(X=x) = \sum_y P(X=x, Y=y)$ and is tabulated in last row of above table.

The marginal probability distribution of Y is given by:

$P(Y=y) = \sum_x P(X=x, Y=y)$ and is tabulated in last column of above table.

(b) The joint probability function:

$$p_{XY}(x, y) = \frac{1}{27}(2x+y); x=0, 1, 2; y=0, 1, 2$$

gives the following table of joint probability distribution of X and Y.

JOINT PROBABILITY DISTRIBUTION $p(x, y)$ OF X AND Y

Y	0	1	2	$f_X(x)$
X				
0	0	$\frac{1}{27}$	$\frac{2}{27}$	$\frac{3}{27}$
1	$\frac{2}{27}$	$\frac{3}{27}$	$\frac{4}{27}$	$\frac{9}{27}$
2	$\frac{4}{27}$	$\frac{5}{27}$	$\frac{6}{27}$	$\frac{15}{27}$

$$\text{For example, } p(0, 0) = \frac{1}{27}(0+2 \times 0) = 0, \quad p(1, 0) = \frac{1}{27}(0+2 \times 1) = \frac{2}{27};$$

$$p(2, 0) = \frac{1}{27}(0+2 \times 2) = \frac{4}{27}; \text{ and so on, } \text{CONDITIONAL DISTRIBUTION OF Y FOR } X=x$$

The conditional distribution of Y for X=x is given by:

$P_{Y|X}(Y=y | X=x) = \frac{p_{XY}(x, y)}{p_X(x)}$ and is obtained in the adjoining table.

X	0	1	2
Y			
0	0	$\frac{1}{3}$	$\frac{2}{3}$
1	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{4}{9}$
2	$\frac{4}{15}$	$\frac{5}{15}$	$\frac{6}{15}$

Example 5.36. Two discrete random variables X and Y have the joint probability density function:

$$p_{XY}(x, y) = \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!}, y=0, 1, 2, \dots, x; x=0, 1, 2, \dots,$$

where λ, p are constants with $\lambda > 0$ and $0 < p < 1$.

Find (i) The marginal probability density functions of X and Y .

(ii) The conditional distribution of Y for a given X and of X for a given Y .

Solution. (i) The marginal p.m.f. of X is given by :

$$\begin{aligned} p_X(x) &= \sum_{y=0}^x p(x, y) = \sum_{y=0}^x \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} = \frac{\lambda^x e^{-\lambda}}{x!} \sum_{y=0}^x \frac{x! p^y (1-p)^{x-y}}{y! (x-y)!} \\ &= \frac{\lambda^x e^{-\lambda}}{x!} \sum_{y=0}^x {}^x C_y p^y (1-p)^{x-y} = \frac{\lambda^x e^{-\lambda}}{x!} [p + (1-p)]^x = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots \end{aligned}$$

which is the probability function of a Poisson distribution with parameter λ .

$$\begin{aligned} p_Y(y) &= \sum_{x=y}^{\infty} p(x, y) = \sum_{x=y}^{\infty} \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} = \frac{(\lambda p)^y e^{-\lambda}}{y!} \sum_{x=y}^{\infty} \frac{[\lambda(1-p)]^{x-y}}{(x-y)!} \\ &[\because y = 0, 1, 2, \dots, x \Rightarrow x \geq y \Rightarrow x - y] \\ &= \frac{(\lambda p)^y e^{-\lambda}}{y!} e^{\lambda(1-p)} = \frac{e^{-\lambda p} (\lambda p)^y}{y!}; \quad y = 0, 1, 2, \dots \end{aligned}$$

which is the probability function of a Poisson distribution with parameter λp .

(ii) The conditional distribution of Y for given X is :

$$\begin{aligned} P_{Y|X}(y | x) &= \frac{p_{XY}(x, y)}{p_X(x)} = \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y} x!}{y! (x-y)! \lambda^x e^{-\lambda}} = \frac{x!}{y! (x-y)!} p^y (1-p)^{x-y} \\ &= {}^x C_y p^y (1-p)^{x-y}, \quad x \geq y \text{ i.e., } y = 0, 1, 2, \dots, x. \end{aligned}$$

The conditional probability distribution of X for given Y is :

$$\begin{aligned} p_{X|Y}(x | y) &= \frac{p_{XY}(x, y)}{p_Y(y)} = \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} \cdot \frac{y!}{e^{-\lambda p} (\lambda p)^y} \\ &= \frac{e^{-\lambda q} (\lambda q)^{x-y}}{(x-y)!}; \quad q = 1-p, \quad x \geq y \text{ i.e., } x = y, y+1, y+2, \dots \end{aligned}$$

Example 5.37. If X and Y are two random variables having joint density function :

$$f(x, y) = \begin{cases} \frac{1}{8}(6-x-y); & 0 \leq x < 2, 2 \leq y < 4 \\ 0, & \text{otherwise} \end{cases}$$

Find (i) $P(X < 1 | Y < 3)$, (ii) $P(X + Y < 3)$, and (iii) $P(X < 1 | Y < 3)$.

Solution. We have

$$\begin{aligned} \text{(i) } P(X < 1 | Y < 3) &= \frac{\int_{-\infty}^1 \int_{-\infty}^3 f(x, y) dx dy}{\int_{-\infty}^1 \int_{-\infty}^3 \frac{1}{8}(6-x-y) dx dy} = \frac{\int_0^1 \int_2^3 \frac{1}{8}(6-x-y) dx dy}{\int_0^1 \int_2^3 \frac{1}{8}(6-x-y) dx dy} = \frac{3}{8} \\ \text{(ii) } P(X + Y < 3) &= \int_0^1 \int_2^{3-x} \frac{1}{8}(6-x-y) dx dy = \frac{5}{24} \\ \text{(iii) } P(X < 1 | Y < 3) &= \frac{P(X < 1 \cap Y < 3)}{P(Y < 3)} = \frac{3/8}{5/8} = \frac{3}{5} \end{aligned}$$

Example 5.38. Suppose that two-dimensional continuous random variable (X, Y) has joint p.d.f. given by :

$$f(x, y) = \begin{cases} 6x^2y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

(i) Verify that $\int_0^1 \int_0^1 f(x, y) dx dy = 1$.

(ii) Find $P(0 < X < \frac{3}{4}, \frac{1}{3} < Y < 2)$, $P(X + Y < 1)$, $P(X > Y)$ and $P(X < 1 | Y < 2)$.

Solution. (i)

$$\int_0^1 \int_0^1 f(x, y) dx dy = \int_0^1 \int_0^1 6x^2y dx dy = \int_0^1 6x^2 \left[\frac{y^2}{2} \right]_0^1 dx = \int_0^1 3x^2 dx = \left[x^3 \right]_0^1 = 1$$

(ii) $P(0 < X < \frac{3}{4}, \frac{1}{3} < Y < 2) = \int_0^{3/4} \int_{1/3}^1 6x^2y dx dy + \int_0^{3/4} \int_1^2 0 dx dy$

$$= \int_0^{3/4} 6x^2 \left[\frac{y^2}{2} \right]_{1/3}^1 dx = \frac{8}{9} \int_0^{3/4} 3x^2 dx = \frac{8}{9} \left[x^3 \right]_0^{3/4} = \frac{3}{8}$$

$$P(X + Y < 1) = \int_0^1 \int_0^{1-x} 6x^2y dx dy = \int_0^1 6x^2 \left[\frac{y^2}{2} \right]_0^{1-x} dx = \int_0^1 3x^2(1-x)^2 dx = \frac{1}{10}$$

$$P(X > Y) = \int_0^1 \int_0^x 6x^2y dx dy = \int_0^1 3x^2 \left[y^2 \right]_0^x dx = \int_0^1 3x^4 dx = \frac{3}{5}$$

$$P(X < 1 | Y < 2) = \frac{P(X < 1 \cap Y < 2)}{P(Y < 2)}$$

$$\text{where } P(X < 1 \cap Y < 2) = \int_0^1 \int_0^1 6x^2y dx dy + \int_0^1 \int_1^2 0 dx dy = 1$$

and $P(Y < 2) = \int_0^1 \int_0^2 f(x, y) dx dy = \int_0^1 \int_0^1 6x^2y dx dy + \int_0^1 \int_1^2 0 dx dy = 1$

$$P(X < 1 | Y < 2) = \frac{P(X < 1 \cap Y < 2)}{P(Y < 2)} = 1.$$

Example 5.39. The joint probability density function of a two-dimensional random variable (X, Y) is given by :

$$f(x, y) = \begin{cases} 2; & 0 < x < 1, 0 < y < x; \\ 0, & \text{elsewhere} \end{cases}$$

(i) Find the marginal density functions of X and Y .

(ii) Find the conditional density function of Y given $X = x$ and conditional density function of X given $Y = y$.

(iii) Check for independence of X and Y .

Solution. Evidently $f(x, y) \geq 0$ and $\int_0^1 \int_0^x 2 dx dy = 2 \int_0^1 x dx = 1$.

(i) The marginal p.d.f.'s of X and Y are given by :

$$f_X(x) = \begin{cases} \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_0^x 2 dy = 2x, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$f_Y(y) = \begin{cases} \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_y^1 2dx = 2(1-y), & 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

(ii) The conditional density function of Y given X , ($0 < x < 1$) is:

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x}, \quad 0 < y < x.$$

The conditional density function of X given Y , ($0 < y < 1$) is:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \frac{2}{2(1-y)} = \frac{1}{(1-y)}, \quad y < x < 1$$

(iii) Since $f_X(x)f_Y(y) = 2(2x)(1-y) \neq f_{XY}(x, y)$, X and Y are not independent.

Example 5.40. The joint p.d.f. of two random variables X and Y is given by:

$$f(x, y) = \frac{9(1+x+y)}{2(1+x)^4(1+y)^4}; \quad 0 \leq x < \infty, 0 \leq y < \infty$$

Find the marginal distributions of X and Y , and the conditional distribution of Y for $X = x$.

Solution. Marginal p.d.f. of X is given by:

$$\begin{aligned} f_X(x) &= \int_0^{\infty} f(x, y) dy = \frac{9}{2(1+x)^4} \int_0^{\infty} \frac{(1+y)+x}{(1+y)^4} dy \\ &= \frac{9}{2(1+x)^4} \int_0^{\infty} \{(1+y)^{-3} + x(1+y)^{-4}\} dy \\ &= \frac{9}{2(1+x)^4} \left(\left| \frac{-1}{2(1+y)^2} \right|_0^{\infty} + x \left| \frac{-1}{3(1+y)^3} \right|_0^{\infty} \right) \\ &= \frac{9}{2(1+x)^4} \cdot \left(\frac{1}{2} + \frac{x}{3} \right) = \frac{3}{4} \cdot \frac{3+2x}{(1+x)^4}; \quad 0 < x < \infty \end{aligned}$$

Since $f(x, y)$ is symmetric in x and y , the marginal p.d.f. of Y is given by:

$$f_Y(y) = \int_0^{\infty} f(x, y) dx = \frac{3}{4} \cdot \frac{3+2y}{(1+y)^4}; \quad 0 < y < \infty$$

The conditional distribution of Y for $X = x$ is given by:

$$f_{Y|X}(Y=y|X=x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{9(1+x+y)}{2(1+x)^4(1+y)^4} \times \frac{4(1+x)^4}{3(3+2x)} = \frac{6(1+x+y)}{(1+y)^4(3+2x)}; \quad 0 < y < \infty$$

Example 5.41. Joint distribution of X and Y is given by:

$$f(x, y) = 4xy e^{-(x^2+y^2)}; \quad x \geq 0, y \geq 0.$$

Test whether X and Y are independent. For the above joint distribution, find the conditional density of X given $Y = y$.

Solution. Joint p.d.f. of X and Y is: $f_{XY}(x, y) = 4xy e^{-(x^2+y^2)}; \quad x \geq 0, y \geq 0$.

Marginal density of X is given by:

$$\begin{aligned} f_X(x) &= \int_0^{\infty} f_{XY}(x, y) dy = \int_0^{\infty} 4xy e^{-(x^2+y^2)} dy = 4x e^{-x^2} \int_0^{\infty} y e^{-y^2} dy \\ &= 4x e^{-x^2} \cdot \int_0^{\infty} e^{-t} \cdot \frac{dt}{2} = 2x \cdot e^{-x^2} \left| -e^{-t} \right|_0^{\infty} \end{aligned}$$

$$\therefore f_X(x) = 2x e^{-x^2}; \quad x \geq 0$$

Similarly, the marginal p.d.f. of Y is given by:

$$f_Y(y) = \int_0^{\infty} f_{XY}(x, y) dx = 2y e^{-y^2}; \quad y \geq 0$$

Since $f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$, X and Y are independently distributed. The conditional distribution of X for given Y is given by:

$$f_{X|Y}(X=x|Y=y) = \frac{f(x, y)}{f_Y(y)} = 2x e^{-x^2}; \quad x \geq 0.$$

Example 5.42. Let X and Y be jointly distributed with p.d.f.:

$$f_{XY}(x, y) = \begin{cases} \frac{1}{4}(1+xy), & |x| < 1, |y| < 1 \\ 0, & \text{otherwise} \end{cases}$$

Show that X and Y are not independent by X^2 and Y^2 are independent.

$$\text{Solution. } f_X(x) = \int_{-1}^1 f(x, y) dy = \frac{1}{4} \left| y + \frac{xy^2}{2} \right|_{-1}^1 = \frac{1}{2}, \quad -1 < x < 1;$$

$$\text{Similarly, } f_Y(y) = \int_{-1}^1 f(x, y) dx = \frac{1}{2}, \quad -1 < y < 1$$

Since $f_{XY}(x, y) \neq f_X(x)f_Y(y)$, X and Y are not independent. However,

$$P(X^2 \leq x) = P(|X| \leq \sqrt{x}) = \int_{-\sqrt{x}}^{\sqrt{x}} f_X(x) dx = \sqrt{x} \quad \dots (*)$$

$$\begin{aligned} P(X^2 \leq x \cap Y^2 \leq y) &= P(|X| \leq \sqrt{x} \cap |Y| \leq \sqrt{y}) \\ &= \int_{-\sqrt{x}}^{\sqrt{x}} \left[\int_{-\sqrt{y}}^{\sqrt{y}} f(u, v) dv \right] du = \sqrt{x} \sqrt{y} \\ &= P(X^2 \leq x) \cdot P(Y^2 \leq y) \end{aligned}$$

[From (*)]

Hence, X^2 and Y^2 are independent.

Example 5.43. A gun is aimed at a certain point (origin of the coordinate system). Because of the random factors, the actual hit point can be any point (X, Y) in a circle of radius R about the origin. Assume that the joint density of X and Y is constant in this circle given by:

$$f_{XY}(x, y) = \begin{cases} k, & \text{for } x^2 + y^2 \leq R^2 \\ 0, & \text{otherwise} \end{cases}$$

(i) Compute k ,

$$(ii) \text{ show that } f_X(x) = \begin{cases} \frac{2}{\pi R} \left\{ 1 - \left(\frac{x}{R} \right)^2 \right\}^{1/2}, & \text{for } -R \leq x \leq R \\ 0, & \text{otherwise} \end{cases}$$

Solution. (i) The constant k is computed from the consideration that the total probability is 1, i.e.,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \Rightarrow \iint_{x^2 + y^2 < R^2} k dx dy = 1 \Rightarrow 4 \int_0^R \int_0^{2\pi} k dx dy = 1$$

where region I is the first quadrant of the circle $x^2 + y^2 = R^2$.

$$\Rightarrow 4k \int_0^R \left(\int_0^{\sqrt{R^2-x^2}} 1 \cdot dy \right) dx = 1 \Rightarrow 4k \int_0^R \sqrt{R^2-x^2} dx = 1$$

$$\Rightarrow 4k \left[x \sqrt{R^2-x^2} + \frac{R^2}{2} \sin^{-1} \left(\frac{x}{R} \right) \right]_0^R = 1 \Rightarrow 4k \left(\frac{R^2}{2} \cdot \frac{\pi}{2} \right) = 1 \Rightarrow k = \frac{1}{\pi R^2}$$

$$\therefore f_{XY}(x, y) = \begin{cases} 1/(\pi R^2); & x^2 + y^2 \leq R^2 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{(ii) } f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \frac{1}{\pi R^2} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} 1 \cdot dy \\ &= \frac{2}{\pi R^2} \int_0^{\sqrt{R^2-x^2}} 1 \cdot y = \frac{2}{\pi R^2} (R^2 - x^2)^{1/2} \\ &= \frac{2}{\pi R} \left\{ 1 - \left(\frac{x}{R} \right)^2 \right\}^{1/2}, \quad -R \leq x \leq R \end{aligned}$$

Example 5.44. Given : $f(x, y) = e^{-(x+y)} I_{(0, \infty)}(x) \cdot I_{(0, \infty)}(y)$. Are X and Y independent?

Find (i) $P(X > 1)$, (ii) $P(X < Y \mid X < 2Y)$, (iii) $P(1 < X + Y < 2)$.

Solution. We are given :

$$f(x, y) = e^{-(x+y)}; 0 \leq x < \infty, 0 \leq y < \infty$$

$$= (e^{-x})(e^{-y}) = f_X(x) \cdot f_Y(y); 0 \leq x < \infty, 0 \leq y < \infty$$

$\Rightarrow X$ and Y are independent and $f_X(x) = e^{-x}; x \geq 0$ and $f_Y(y) = e^{-y}; y \geq 0$

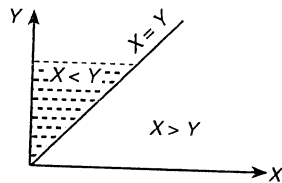
$$\begin{aligned} \text{(i) } P(X > 1) &= \int_1^{\infty} f_X(x) dx \\ &= \int_1^{\infty} e^{-x} dx = \left[\frac{e^{-x}}{-1} \right]_1^{\infty} = \frac{1}{e} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(X < Y \mid X < 2Y) &= \frac{P(X < Y \cap X < 2Y)}{P(X < 2Y)} \\ &= \frac{P(X < Y)}{P(X < 2Y)} \quad \dots (3) \end{aligned}$$

$$\begin{aligned} P(X < Y) &= \int_0^{\infty} \left\{ \int_0^y f(x, y) dx \right\} dy = \int_0^{\infty} \left\{ e^{-y} \left[\frac{e^{-x}}{-1} \right]_0^y \right\} dy \\ &= - \int_0^{\infty} e^{-y} (e^{-y} - 1) dy = \left[\frac{e^{-2y}}{-2} + e^{-y} \right]_0^{\infty} = 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} P(X < 2Y) &= \int_0^{\infty} \left\{ \int_0^{2y} f(x, y) dx \right\} dy = - \int_0^{\infty} e^{-y} (e^{-2y} - 1) dy \\ &= - \left[\frac{e^{-3y}}{-3} + e^{-y} \right]_0^{\infty} = 1 - \frac{1}{3} = \frac{2}{3} \end{aligned}$$

Substituting in (3), $P(X < Y \mid X < 2Y) = \frac{1/2}{2/3} = \frac{3}{4}$



$$\text{(iii) } P(1 < X + Y < 2)$$

$$= \iint_I f(x, y) dx dy$$

$$+ \iint_{II} f(x, y) dx dy$$

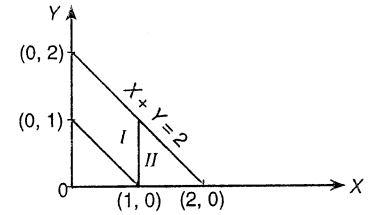
$$= \int_0^1 \left(\int_{1-x}^{2-x} f(x, y) dy \right) dx + \int_1^2 \left(\int_0^{2-x} f(x, y) dy \right) dx$$

$$= \int_0^1 \left(e^{-x} \int_{1-x}^{2-x} e^{-y} dy \right) dx + \int_1^2 \left(e^{-x} \int_0^{2-x} e^{-y} dy \right) dx$$

$$= \int_0^1 \frac{e^{-x}}{-1} (e^{x-2} - e^{x-1}) dx + \int_1^2 \frac{e^{-x}}{-1} (e^{x-2} - 1) dx$$

$$= -(e^{-2} - e^{-1}) \int_0^1 1 \cdot dx - \int_1^2 (e^{-2} - e^{-x}) dx$$

$$= -(e^{-2} - e^{-1}) \left[x \right]_0^1 - \left[e^{-2} \cdot x + e^{-x} \right]_1^2 = \frac{2}{e} - \frac{3}{e^2}$$



Example 5.45. If the joint distribution function of X and Y is given by :

$$F_{XY}(x, y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-(x+y)}; & x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Find the marginal densities of X and Y .

(b) Are X and Y independent?

(c) Find $P(X \leq 1 \cap Y \leq 1)$ and $P(X + Y \leq 1)$.

Solution. (a) & (b). The joint p.d.f. of the r.v.'s (X, Y) is given by :

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y} = \frac{\partial}{\partial x} [e^{-y} - e^{-(x+y)}] = \begin{cases} e^{-(x+y)}; & x \geq 0, y \geq 0 \\ 0; & \text{otherwise} \end{cases} \quad \dots (i)$$

We have $f_{XY}(x, y) = e^{-x} \cdot e^{-y} = f_X(x) f_Y(y)$

$\dots (ii)$

where $f_X(x) = e^{-x}; x \geq 0; f_Y(y) = e^{-y}; y \geq 0$

$\dots (iii)$

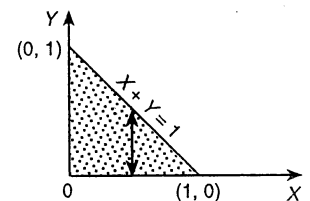
(ii) $\Rightarrow X$ and Y are independent, and (iii) gives the marginal p.d.f.'s of X and Y .

$$\begin{aligned} \text{(c) } P(X \leq 1 \cap Y \leq 1) &= \int_0^1 \int_0^1 f(x, y) dx dy \\ &= \left(\int_0^1 e^{-x} dx \right) \left(\int_0^1 e^{-y} dy \right) = (1 - e^{-1})^2 \end{aligned}$$

$$P(X + Y \leq 1) = \iint_{X+Y \leq 1} f(x, y) = \int_0^1 \left\{ \int_0^{1-x} f(x, y) dy \right\} dx$$

$$= \int_0^1 \left[e^{-x} \int_0^{1-x} e^{-y} dy \right] dx$$

$$= \int_0^1 e^{-x} \{ 1 - e^{-(1-x)} \} dx = 1 - 2e^{-1}$$



Example 5-46. (i) Let $F(x, y)$ be the d.f. of X and Y . Show that

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(b, c) - F(a, d) + F(a, c),$$

where a, b, c and d are real constants $a < b; c < d$.

$$\text{Deduce that if: } F(x, y) = \begin{cases} 1, & \text{for } x + 2y \geq 1 \\ 0, & \text{for } x + 2y < 1, \end{cases}$$

then $F(x, y)$ cannot be joint distribution function of variables X and Y .

(ii) Show that, with usual notations: for all x, y ,

$$F_X(x) + F_Y(y) - 1 \leq F_{XY}(x, y) \leq \sqrt{F_X(x) F_Y(y)}.$$

Solution. (i) Let us define the events:

$$A: \{X \leq a\}; \quad B: \{X \leq b\}; \quad C: \{Y \leq c\}; \quad D: \{Y \leq d\}; \quad \text{for } a < b; c < d$$

$$P(a < X \leq b \cap c < Y \leq d)$$

$$= P[(B - A) \cap (D - C)]$$

$$= P[B \cap (D - C) - A \cap (D - C)] \quad \dots (*)$$

(By distributive property of sets)

We know that if $E \subset F \Rightarrow E \cap F = E$, then

$$P(F - E) = P(\bar{E} \cap F) = P(F) - P(E \cap F) = P(F) - P(E)$$

$$\text{Obviously } A \subset B \Rightarrow [A \cap (D - C)] \subset [B \cap (D - C)]$$

Hence using (**), we get from (*)

$$\begin{aligned} P(a < X \leq b \cap c < Y \leq d) &= P[B \cap (D - C)] - P[A \cap (D - C)] \\ &= P[(B \cap D) - (B \cap C)] - P[(A \cap D) - (A \cap C)] \\ &= P(B \cap D) - P(B \cap C) - P(A \cap D) + P(A \cap C) \quad \dots (***) \end{aligned}$$

[On using (**), since $C \subset D \Rightarrow (B \cap C) \subset (B \cap D)$ and $(A \cap C) \subset (A \cap D)$]

$$\text{We have: } P(B \cap D) = P[X \leq b \cap Y \leq d] = F(b, d).$$

$$\text{Similarly } P(B \cap C) = F(b, c); \quad P(A \cap D) = F(a, d) \text{ and } P(A \cap C) = F(a, c)$$

Substituting in (***), we get

$$P(a < X \leq b \cap c < Y \leq d) = F(b, d) - F(b, c) - F(a, d) + F(a, c) \quad \dots (1)$$

$$\text{We are given } F(x, y) = \begin{cases} 1, & \text{for } x + 2y \geq 1 \\ 0, & \text{for } x + 2y < 1 \end{cases} \quad \dots (2)$$

In (1) let us take $a = 0, b = \frac{1}{2}; c = \frac{1}{4}, d = \frac{3}{4}$ s.t. $a < b$ and $c < d$. Then using (2), we get

$$F(b, d) = 1; \quad F(b, c) = 1; \quad F(a, d) = 1; \quad F(a, c) = 0.$$

Substituting in (1), $P(a < X \leq b \cap c < Y \leq d) = 1 - 1 - 1 + 0 = -1$; which is not possible since $P(\cdot) \geq 0$.

Hence $F(x, y)$ defined in (2) cannot be the distribution function of r.v.'s X and Y .

(ii) Let us define the events: $A: \{X \leq x\}; \quad B: \{Y \leq y\}$

$$\text{Then: } P(A) = P(X \leq x) = F_X(x); \quad P(B) = P(Y \leq y) = F_Y(y) \quad \dots (3)$$

$$\text{and } P(A \cap B) = P(X \leq x \cap Y \leq y) = F_{XY}(x, y)$$

$$(A \cap B) \subset A \Rightarrow P(A \cap B) \leq P(A) \Rightarrow F_{XY}(x, y) \leq F_X(x)$$

$$(A \cap B) \subset B \Rightarrow P(A \cap B) \leq P(B) \Rightarrow F_{XY}(x, y) \leq F_Y(y)$$

Multiplying these inequalities, we get

$$F_{XY}^2(x, y) \leq F_X(x) F_Y(y) \Rightarrow F_{XY}(x, y) \leq \sqrt{F_X(x) F_Y(y)} \quad \dots (4)$$

$$\text{Also } P(A \cup B) \leq 1 \Rightarrow P(A) + P(B) - P(A \cap B) \leq 1 \Rightarrow P(A) + P(B) - 1 \leq P(A \cap B)$$

$$\therefore F_X(x) + F_Y(y) - 1 \leq F_{XY}(x, y) \quad \dots (5)$$

From (4) and (5), $F_X(x) + F_Y(y) - 1 \leq F_{XY}(x, y) \leq \sqrt{F_X(x) F_Y(y)}$, as required.

5.6. TRANSFORMATION OF ONE-DIMENSIONAL RANDOM VARIABLE

Let X be a r.v. defined on the event space S and let $g(\cdot)$ be a function such that $Y = g(X)$ is also a r.v. defined on S . In this section we shall deal with the following problem:

"Given the probability density function of a r.v. X , to determine the probability density function of a new r.v. $Y = g(X)$."

It can be proved in general that, if $g(\cdot)$ is any continuous function, then the distribution of $Y = g(X)$ is uniquely determined by that of X . The proof of this result is rather difficult and beyond the scope of this book. Here, we shall consider the following, relatively simple theorem.

Theorem 5-4. Let X be a continuous r.v. with p.d.f. $f_X(x)$. Let $y = g(x)$ be strictly monotonic (increasing or decreasing) function of x . Assume that $g(x)$ is differentiable (and hence continuous) for all x . Then the p.d.f. $h_Y(y)$ of the r.v. Y is given by:

$$h_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|, \quad \dots (5-22)$$

where x is expressed in terms of y , and the range of Y is determined from the given range of the variable X , on using the transformation $y = g(x)$.

Proof. Case (i). $y = g(x)$ is strictly increasing function of x (i.e., $\frac{dy}{dx} > 0$). The d.f. of Y is given by:

$$H_Y(y) = P(Y \leq y) = P\{g(X) \leq y\} = P\{X \leq g^{-1}(y)\},$$

the inverse exists and is unique, since $g(\cdot)$ is strictly increasing.

$$\therefore H_Y(y) = F_X(g^{-1}(y)), \text{ where } F \text{ is the d.f. of } X$$

$$= F_X(x)$$

$$[\because y = g(x) \Rightarrow g^{-1}(y) = x]$$

Differentiating w.r. to y , we get

$$h_Y(y) = \frac{d}{dy} [F_X(x)] = \frac{d}{dx} [F_X(x)] \frac{dx}{dy} = f_X(x) \frac{dx}{dy} \quad \dots (*)$$

Case (ii). $y = g(x)$ is strictly monotonic decreasing function of x .

$$\begin{aligned} H_Y(y) &= P(Y \leq y) = P\{g(X) \leq y\} = P\{X \geq g^{-1}(y)\} \\ &= 1 - P\{X \leq g^{-1}(y)\} = 1 - F_X(g^{-1}(y)) = 1 - F_X(x), \end{aligned}$$

where $x = g^{-1}(y)$, the inverse exists and is unique. Differentiating w.r. to y , we get

$$h_Y(y) = \frac{d}{dx} [1 - F_X(x)] \frac{dx}{dy} = -f_X(x) \frac{dx}{dy} = f_X(x) \frac{-dx}{dy} \quad \dots (**)$$

Note that the algebraic sign (-ve) obtained in (**) is correct, since y is a decreasing function of $x \Rightarrow x$ is a decreasing function of y or $\frac{dx}{dy} < 0$.

The results in (*) and (**) can be combined to give $h_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$.

Example 5.47. If the cumulative distribution function of a continuous r.v. X is $F(x)$ find the cumulative distribution function of:

- (i) $Y = X + a$, (ii) $Y = X - b$, (iii) $Y = aX$, (iv) $Y = X^3$, and (v) $Y = X^2$.

What are the corresponding probability density functions?

Solution. Let $G(\cdot)$ be the c.d.f. of Y . Then

$$(i) \quad G(x) = P(Y \leq x) = P(X + a \leq x) = P(X \leq x - a) = F(x - a)$$

$$(ii) \quad G(x) = P(Y \leq x) = P(X - b \leq x) = P(X \leq x + b) = F(x + b)$$

$$(iii) \quad G(x) = P(aX \leq x) = P\{X \leq (x/a)\} = F(x/a), \quad \text{if } a > 0$$

$$\text{and} \quad G(x) = P\{X \geq (x/a)\} = 1 - P\{X \leq (x/a)\} = 1 - F(x/a), \quad \text{if } a < 0.$$

$$(iv) \quad G(x) = P(Y \leq x) = P(X^3 \leq x) = P(X \leq x^{1/3}) = F(x^{1/3})$$

$$(v) \quad G(x) = P(X^2 \leq x) = P(-x^{1/2} \leq X \leq x^{1/2}) = P(X \leq x^{1/2}) - P(X \leq -x^{1/2})$$

$$= \begin{cases} 0 & , \text{ if } x < 0 \\ F(\sqrt{x}) - F(-\sqrt{x}) & , \text{ if } x > 0 \end{cases}$$

Variable	Distribution function	p.d.f.
X	$F(x)$	$f(x)$
$X - a$	$F(x + a)$	$f(x + a)$
aX	$\begin{cases} F(x/a), & a > 0 \\ 1 - F(x/a), & a < 0 \end{cases}$	$\begin{cases} \frac{1}{a} f(x/a), & a > 0 \\ -\frac{1}{a} f(x/a), & a < 0 \end{cases}$
X^2	$\begin{cases} F(\sqrt{x}) - F(-\sqrt{x}), & \text{for } x > 0 \\ 0, & \text{otherwise} \end{cases}$	$\begin{cases} \frac{1}{2\sqrt{x}} [f(\sqrt{x}) + f(-\sqrt{x})], & \text{for } x > 0 \\ 0, & \text{for } x \leq 0 \end{cases}$
X^3	$F(x^{1/3})$	$\frac{1}{3} f(x^{1/3}) \cdot \frac{1}{x^{2/3}}$

5.7. TRANSFORMATION OF TWO-DIMENSIONAL RANDOM VARIABLE

In this section we shall consider the problem of change of variables in the two-dimensional case. Let r.v.'s U and V be transformed to the r.v.'s X and Y by the transformation $u = u(x, y)$, $v = v(x, y)$, where u and v are continuously differentiable

functions, for which Jacobian of transformation: $J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix}$, ... (5.23)

is either > 0 or < 0 throughout the (x, y) plane so that the inverse transformation is uniquely given by $x = x(u, v)$, $y = y(u, v)$.

Theorem 5.5. The joint p.d.f. $g_{UV}(u, v)$ of the transformed variables U and V is: $g_{UV}(u, v) = f_{XY}(x, y) |J|$, ... (5.24)

where $|J|$ is the modulus value of the Jacobian of transformation and $f(x, y)$ is expressed in terms of u and v .

Proof. $P(x < X \leq x + dx, y < Y \leq y + dy) = P(u < U \leq u + du, v < V \leq v + dv)$
 $\Rightarrow f_{XY}(x, y) dx dy = g_{UV}(u, v) du dv$

$$\Rightarrow g_{UV}(u, v) du dv = f_{XY}(x, y) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

$$\therefore g_{UV}(u, v) = f_{XY}(x, y) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = f_{XY}(x, y) |J|$$

Theorem 5.6. (a) If X and Y are independent continuous r.v.'s, then the p.d.f. of $U = X + Y$ is given by: $h(u) = \int_{-\infty}^{\infty} f_X(v) f_Y(u - v) dv$... (5.25)

Proof. Let $f_{XY}(x, y)$ be the joint p.d.f. of independent continuous r.v.'s X and Y and let us make the transformation: $u = x + y, v = x \Rightarrow x = v, y = u - v$.

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

Thus the joint p.d.f. of r.v.'s U and V is given by:

$$g_{UV}(u, v) = f_{XY}(x, y) |J| = f_X(x) f_Y(y) |J| \quad (\text{Since } X \text{ and } Y \text{ are independent})$$

$$= f_X(v) f_Y(u - v)$$

The marginal density of U is given by:

$$h(u) = \int_{-\infty}^{\infty} g_{UV}(u, v) dv = \int_{-\infty}^{\infty} f_X(v) f_Y(u - v) dv$$

Remark. The function $h(\cdot)$ is given a special name and is said to be the *convolution* of $f_X(\cdot)$ and $f_Y(\cdot)$ and we write $h(\cdot) = f_X(\cdot) * f_Y(\cdot)$.

Alter. The distribution function $G(\cdot)$ of $U = X + Y$, is given by:

$$G_U(u) = P(U \leq u) = \iint_A f(x, y) dx dy,$$

where $A = \{(x, y) : x + y \leq u\}$. Hence on $A : y \leq u - x$.

$$\therefore G_U(u) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{u-x} f(x, y) dy \right) dx$$

$$= \int_{-\infty}^{\infty} \left(\int_{-\infty}^u f(x, z - x) dz \right) dx \quad [x + y = z \Rightarrow dy = dz]$$

$$= \int_{-\infty}^u \left(\int_{-\infty}^{\infty} f(x, z - x) dx \right) dz$$

(By Fubini's Theorem for non-negative functions)

Differentiating w.r. to u , the p.d.f. of $U = X + Y$, is given by:

$$g(u) = \int_{-\infty}^{\infty} f(x, u - x) dx \quad \dots (5.26)$$

If X and Y are independent, then

$$g(u) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(u - x) dx, \quad \dots (5.26a)$$

an expression obtained in (5.25).